

AQbD Analytical Method Development Report

Analytical Quality by Design (AQbD) / Design of Experiments (DOE) Studio V1

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Project Name:	HPLC Assay & Impurity Method Development	Software:	AQbD Studio V1 (ICH Q14 Enhanced Approach)

1. Executive Summary

This report documents the systematic, science- and risk-based development of the analytical procedure following ICH Q14 guidelines. The workflow progresses from defining the Analytical Target Profile (ATP) and initial Failure Mode and Effects Analysis (FMEA) risk assessment, to multivariate Design of Experiments (DOE), rigorous ANOVA and diagnostic model verification, multi-response desirability optimization, design space boundaries determination, and experimental confirmation.

2. Analytical Target Profile (ATP)

Parameter	Criterion	Unit	Acceptance Limits / Target
Assay Accuracy	RANGE	%	[98.0 to 102.0]

3. Risk Assessment (FMEA)

Parameter	Severity (S)	Occurrence (O)	Detectability (D)	RPN	Priority
Column Temperature	4	3	2	24	Medium

4. Experimental Factors (DOE Parameters)

Code	Factor Name	Unit	Low (-1)	High (+1)	Role
A	Flow Rate	mL/min	0.8	1.2	CRITICAL
B	Column Temperature	°C	30.0	40.0	CRITICAL

5. Experimental Responses & Goals

Code	Response Name	Unit	Goal	Limits / Target	Weight
Y1	Resolution	Rs	MAXIMIZE	-	5
Y2	Retention Time	min	TARGET	= 6.0	4

6. Design of Experiments (DOE) Execution

Design Type: CCD | Runs: 12 | Random Seed: 42

7. Statistical Modeling & Optimization

Model: QUADRATIC | R²: N/A | Adj R²: N/A | Pred R²: N/A
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Multi-Response Optimal Solution (Desirability):

Overall Desirability: **0.6463**

Factor Settings: A = 0.800, B = 39.464

8. Experimental Confirmation

Predicted: {'Y1': 2.438633098863857, 'Y2': 5.99587447262894} | Actual: {'Y1': 2.459, 'Y2': 5.946} | Difference: {'Y1': 0.020366901136143145, 'Y2': 0.0498744726289404}

9. Conclusion

The analytical method was developed and evaluated using systematic AQbD principles. The mathematical models, design space boundaries, and confirmation verification prove that the procedure operates robustly within its acceptable ranges.

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